

Roll no.

END SEMESTER EXAMINATION 2025

Name of the Course: BTech

Semester: 8

Name of the Paper: Unix Systems Programming

Paper code: TCS 826

Time: 3 Hours

Maximum Marks: 100

Note:-

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main question.
- iii. Total marks in each main question are 20.
- iv. Each question carries 10 marks.

Q1.

- a) With a suitable code example, explain how a shell runs a command in Unix. [CO4]
- b) Write a program that illustrates the system calls used to first display the priority of a process and then to modify the priority of the process. [CO5]
- c) Write a program *myls* whose output is similar to the *ls* command that displays the list of files in a directory `"/home/cse/"`. Don't use the *ls* command in your program. [CO3]

Q2

- a) With an example, illustrate the use of `nice()` system. [CO6]
- b) Differentiate between normal process scheduling and real-time process scheduling. Give example of real-time scheduling policies in Linux. [CO6]
- c) With the help of a program, illustrate the use of `msgsnd()` and `msgrcv()` system calls in message queue. [CO2]

Q3.

- a) With suitable examples, differentiate between reliable and unreliable signals. [CO1]
- b) With the help of suitable examples, explain the difference between buffered I/O and unbuffered I/O. [CO1]
- c) Illustrate the problem with `dup()` that is solved by the `dup2()` system call. [CO1]

Q4.

- a) With an example, explain the use of signals and signal handlers in Unix. [CO3]
- b) Create a Signal Handler using C/Unix programming that prints "Caught it" whenever you try to terminate the process by pressing Ctrl-C. [CO3]
- c) What is signal mask? Write the names of two signals that cannot be blocked. [CO3]

Q5.

- a) Using semaphores and mutex, write code for producer/consumer using semaphore and mutex. There are 5 producers that write to a circular buffer of size 5 and 5 consumers that read from the buffer. [CO2]
- b) Write a small program to illustrate the client server programming in Unix. [CO5]
- c) Discuss the design issues in concurrent and iterative servers. [CO5]

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